Higher National Diploma in Accountancy
First Year, First Semester Examination - 2019
HNDA 1102 - Business Mathematics

Instructions for candidates:	No of Questions :06
Answer **five (5) questions**	No of Pages :04
Calculators are Allowed	Time :**Three hours**

Question 01

(i) Solve $\frac{3x-2}{6} + \frac{2x-1}{4} - \frac{2x}{3} = 0$. (03 Marks)

(ii) Find factors of $3x^2 - x - 14$. (02 Marks)

(iii) If $\log_3 x = -3$ then find the value of x . (02 Marks)

(iii) find the equation of straight line which passes through (1,3) and parallel to 2y = x – 5. (03 Marks)

(v) Find ${}^5C_2 - {}^3P_2$. (02 Marks)

(vi) Solve x (4x - 5) = 6 . (03 Marks)

(vii) Two numbers are chosen such that twice the first plus the second is 17 and four times the first minus the second is 25. What are the two numbers? (03 Marks)

(viii) In an arithmetic progression the 10th term is 125 and common difference is 10, find first term. (02 Marts)

[Total 20 Marks]

Question 02

(i) Suppose there are 5 males and 4 females in a department. We have to select 5 employees to form a team for a project. find out the number of possible that we can make out of 5 employees,

(a) At least 4 males

(b) 2 males and 3 females

(c) How many ways can arrange 10 books on a bookshelf that has space for only 8 books? (4 Marks)

(ii) The first year profit is Rs.25,000 for a business, if the profit of next year equals to 1.2 times of profit of last year. Find the following

(a) The profit expected in the 8th year.
(b) The total profit during first 4 years.
(c) The total profit during second 4 years. (10 Marks)

[Total 20 Marks]

Question 03

(i) Let A be the set of even numbers less than 18, B be the set of multiples of 3 less than 20 and the universal set be the natural numbers less than 20. Find the following.

(a) $n(A \cap B)$
(b) $A^{/}$ (complement of A)
(c) $A^{/} \cap B^{/}$ (07 Marks)

(ii) A class of 100 students analyzed the subjects they like. The subjects are Accounting, Economics and Business Studies. 30 students like all three subjects. 5 students don't like any of these subjects. 40 students like Accounting and Economies. 15 students like Accounting and Business Studies only. 58 students like Economies. 62 students like Business Studies and 66 students like Accounting.

(a) Draw the Venn diagram for above information

(b) How many students like Economies and Business Studies, but not Accounting?

(c) How many students like Business Studies only?

(d) How many students like only two subjects? (13 Marks)

[Total 20 Marks]

Question 04

(i) If Rs. 20 000 is borrowed for 6 months at an annual interest rate of 12%, find the total interest and the amount of the loan at the end of six months. (04 Marks)

(ii) If we invest Rs. 50 000 at the rate of 10% per year compounded semi-annually. How much will we have at the beginning of the 5th years? (06 Marks)

(iii)The following table represents cash flow of two projects:

hndapastpapers.blogspot.com

Year	0	1	2	3	4	5
Cash flow (Project X)	-150,000	60,000	45,000	40,000	39,000	44,000
Cash flow (Project Y)	-250,000	103,000	70,500	59,000	54,100	60,600

Calculate NPV for each project when the rate of return is 15% and find which project is profitable.

(10 Marks)

[Total 20 Marks]

Question 05

(i) Evaluate $\lim_{x \to 1} \frac{x^{50} - 1}{x^{25} - 1}$ (04 Marks)

(ii) Differentiate the following functions:

(a) $Y = (6x^4 - 5x)^2$ (04 Marks)
(b) $Y = (5x - 3)(7x^2 + 5)$ (04 Marks

(iii)Integrate the following functions:

(a) $\int (2x^2 - 4x - 3)\, dx$ (04 Marks)

(b) $\int_3^4 (3x2 + 12x - 4)\, dx$ (04 Marks)

[Total 20 Marks]

Question 06

In a certain production price function is $p = 1000 - 2q$ and the cost function is given by $C = 800 + 10q + 3q^2$, where p is price q is the production level.

Find the following:

(i) Marginal Cost Function (03Marks)

(ii) Marginal Revenue Function (03 Marks)

(iii) Profit function (03 Marks)

(iv) Breakeven points (04 Marks)

(v) Maximum profit (04 Marks)

(vi) Calculate total profit when producing 20 units (03 Marks)

[Total 20 Marks]